

ORIGINAL PILOT STUDY

Machine Learning for Next Day Extreme Rainfall Early Warning in Bangladesh

A multi-city pilot study and rainfall-based flood-risk proxy

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Abstract

This study evaluates machine-learning models for next-day extreme-rainfall warning at eight Bangladeshi divisional cities. Daily NASA POWER observations from 2014-2024 produced 32,032 lag-complete samples. City-specific extreme rainfall was defined above the 95th percentile of 2014-2021 precipitation. Models trained through 2021 and were tested on 2022-2024, where 402 of 8,760 cases were positive. Logistic regression achieved the highest test PR-AUC, 0.348, and recalled 78.1 percent of extreme events, but precision was 19.3 percent. Random forest and histogram boosting produced higher precision but missed most positive cases at a 0.5 threshold. A persistence baseline remained competitive. The results show that rare-event evaluation and decision thresholds matter more than headline accuracy. Rainfall alone is not a flood forecast; the model is a hydroclimate screening tool that requires river level, soil moisture, drainage, and inundation data before operational use.

Research Questions

- Can recent rainfall and meteorological conditions rank next-day extreme-rainfall risk across multiple cities?
- Do nonlinear models outperform a persistence baseline under a future-year test split?
- How do recall and false-alert burden change the preferred model?

Introduction

Heavy rainfall is one input to flooding, but flood impacts also depend on river flow, antecedent soil moisture, drainage, topography, tide, and exposure. Many machine-learning papers report high accuracy on imbalanced hydrological events even when a model fails to identify the rare cases that matter. This study uses PR-AUC, recall, precision, and calibration to evaluate a limited but reproducible next-day warning task.

Related Work

Reviews of flood prediction report frequent use of neural networks, support-vector methods, trees, and hybrid systems [1]. Large-sample hydrology studies show that recurrent networks can learn useful rainfall-runoff relationships when discharge observations are available [2]. WeatherBench and related work emphasize chronological evaluation rather than random splits [3]. The present study follows that evaluation principle but uses precipitation extremes because consistent public river-level labels were not available in the computational package.

Data and Methods

Observations and Labels

NASA POWER supplied daily corrected precipitation, temperature, relative humidity, and wind speed for the eight divisional cities from 2014-2024. The next day's precipitation was compared with each city's 95th-percentile threshold estimated only from 2014-2021. The analytical table contained 23,272 training cases and 8,760 test cases. The test event rate was 4.59 percent.

Predictors and Models

Predictors included current rainfall, seven daily lags, rolling rainfall totals and means over 3, 7, and 14 days, current and lagged temperature, humidity and wind, cyclic day-of-year variables, and city. The

study compared a rainfall persistence score, balanced logistic regression, a 500-tree balanced random forest, and histogram gradient boosting.

Evaluation

PR-AUC was the primary metric because positive events were rare. ROC-AUC measured ranking over both classes; precision and recall described false-alert and missed-event trade-offs; F1 summarized their harmonic mean; balanced accuracy gave equal weight to both classes; and Brier score measured probability error. A fixed 0.5 threshold was retained for an auditable comparison and was not tuned on the test period.

Results

Model	ROC-AUC	PR-AUC	F1	Precision	Recall	Bal Acc	Brier
Logistic regression	0.900	0.348	0.310	0.193	0.781	0.812	0.116
Histogram boosting	0.908	0.337	0.146	0.449	0.087	0.541	0.035
Random forest	0.903	0.331	0.226	0.492	0.147	0.570	0.036
Persistence baseline	0.905	0.305	0.371	0.254	0.689	0.796	0.088

Logistic regression improved PR-AUC by 0.042 over the persistence baseline and produced the highest recall, but roughly four of five alerts were false positives at the default threshold. Random forest reached 0.492 precision but only 0.147 recall. Histogram boosting had the best ROC-AUC and Brier score, yet its 0.087 recall made the default threshold unsuitable for an early-warning objective.

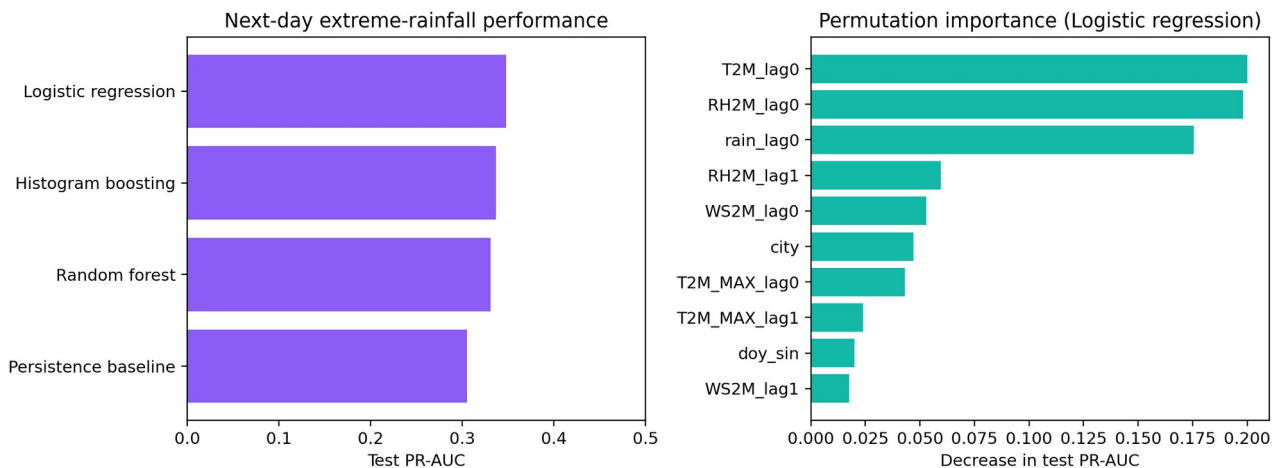


Figure 1 Test PR-AUC and permutation importance. The importance plot is calculated for logistic regression using held-out 2022-2024 observations.

The most influential held-out features were T2M_lag0, RH2M_lag0, rain_lag0, RH2M_lag1, WS2M_lag0, city, T2M_MAX_lag0. The result indicates that current temperature, humidity, rainfall, wind, location, and recent conditions jointly affect the ranking. It does not show that these predictors cause flooding.

Discussion

The pilot reveals why an apparently strong ROC-AUC can be misleading for rare warnings. Logistic regression retrieved most extreme-rain days but generated many false alerts. Tree models produced better precision and probability error but missed events at the fixed threshold. An operational study should set thresholds from explicit warning costs, calibrate probabilities on a separate validation period, and evaluate lead times longer than one day.

Limitations and Research Integrity

- The outcome is extreme rainfall, not measured river flooding, urban waterlogging, inundation depth, or damage.
- Point-scale gridded climate estimates may differ from rain-gauge observations, particularly during localized convection.
- The study does not include upstream discharge, river stage, soil moisture, tide, topography, or drainage capacity.
- The 95th-percentile definition is useful for comparison but is not an official warning threshold.
- Results are a completed pilot experiment and must not be described as operational flood-forecast performance.

Conclusion

Balanced logistic regression achieved a test PR-AUC of 0.348 and recall of 0.781 for next-day extreme rainfall. The result supports further work, but it also quantifies a large false-alert burden. A defensible flood-warning paper should next integrate Bangladesh Water Development Board river levels and independent gauge rainfall, then test by basin and flood season.

Data and Code Availability

The computational package accompanying this manuscript contains the exact Python script, cached source data, derived features, result tables, and figures used in the analysis. The raw climate observations were requested from the NASA POWER Daily API. The machine-learning benchmark datasets were loaded through scikit-learn without manual modification.

Component	Included files
Code	run_experiments.py
Data	NASA POWER city CSV files and paper2_daily_features.csv
Results	paper2_metrics.csv, paper2_permutation_importance.csv, and Figure 1

References

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